

Spray Chain

Decentralized Money Spraying Protocol

Whitepaper v1.0 · September 2026 · FaucetDrops

Celo Mainnet · Botchain (Chain ID 677)

SECTION 00

Abstract

SprayChain is a Web3 celebration protocol that brings the beloved Nigerian and African party tradition of "money spraying" onto the blockchain. Guests at events, fans of artists, or anyone celebrating a person can connect their wallet and spray crypto tokens at an honoree in real time — with a cinematic spray animation, configurable drip rate, and on-chain settlement.

SprayChain turns a cultural ritual into a transparent, peer-to-peer gifting primitive. The sprayer sets the rate, presses the Spray button, and digital dollars fly outward on screen while an on-chain contract accumulates the total. When the session ends, the contract transfers the full amount to the honoree with a human-readable notification: "You received \$12.50 from Akin — Happy Birthday Queen 🍀"

SECTION 01

Background & Motivation

Money spraying is a celebratory tradition common at Nigerian parties, weddings, and festivals. Well-wishers physically rain banknotes on a celebrant as a gesture of honor and abundance. The practice is emotionally resonant and communally visible — everyone sees who is celebrated and how generously.

Web3 gifts today are cold: a wallet-to-wallet transfer, a block explorer hash. SprayChain restores the spectacle. A sprayer selects the rate, presses the Spray button, and the contract accumulates the total in real time before pushing the full balance to the honoree on session close.

SECTION 02

System Architecture

2.1 Component Overview

SprayChain consists of three layers: the browser frontend, the SprayMachine smart contract, and an optional off-chain notification indexer.

```
Sprayer Browser
|
+-- HTML/JS Frontend
|   +-- Wallet connect (MetaMask / wagmi)
|   +-- Party form (name, address, rate)
|   +-- Spray canvas (animated dollar shower)
|   +-- Session summary + receipt
|
v
SprayMachine.sol (Celo Mainnet / Botchain)
+-- openSpray(honoreeAddr, honoreeTag) payable
```

```
+-- addSpray(sessionId) payable
+-- closeSpray(sessionId)
+-- getSession(sessionId) view
|
v
Off-chain Indexer (FastAPI + The Graph)
+-- SprayClosed event listener
+-- WebSocket push to honoree browser
+-- Supabase receipt log
```

SECTION 03

Smart Contract Specification

3.1 SprayMachine.sol — Function Reference

Function	Visibility	Description
openSpray(address, string)	external payable	Opens session, deposits initial amount, emits S
addSpray(bytes32)	external payable	Adds to active session per UI tick
closeSpray(bytes32)	external	Caller closes session; contract pushes balance
getSession(bytes32)	external view	Returns session: sprayer, honoree, tag, total, st
getSprayerHistory(address)	external view	Returns all session IDs opened by a sprayer
getHonoreeHistory(address)	external view	Returns all session IDs received by an honoree

3.2 Events

Event	Indexed Fields	Purpose
SprayOpened	sessionId, sprayer, honoree	Fired when a new session is created
SprayTick	sessionId	Fired on every addSpray() deposit
SprayClosed	sessionId, sprayer, honoree	Fired on closeSpray(); triggers notification

3.3 Security Properties

- Non-custodial: Contract never holds funds across blocks beyond the active session.
- Session ownership: Only the original sprayer may call closeSpray() on their session.
- Reentrancy guard: All state mutations happen before external calls (checks-effects-interactions).
- No admin keys: No owner, pauser, or upgrade proxy — the contract is fully immutable after deployment.

SECTION 04

Token Support

SprayChain v1 launches on Celo Mainnet using native CELO as the spray token (low gas, mobile-first chain). A v1.1 upgrade will add cUSD (Celo Dollar stablecoin) and USDC via ERC-20 approve/transferFrom, making spray amounts denominated in real dollars — natural for the gifting use case.

On Botchain (chain ID 677) the native gas token is used directly, following the same non-custodial model. No bridging or wrapping is required within a single chain.

SECTION 05

Rate Ticks — UX to Contract Mapping

The spray rate chosen in the UI maps to smart contract calls every 500ms. Gas batching groups up to 10 ticks per transaction to keep overhead manageable while the visual animation runs at full speed.

Rate Setting	Tick Value	Contract Behaviour
\$0.10 / second	0.05 USD per tick	addSpray() every 500ms with 0.05 USD equivalent in CELO
\$0.50 / second	0.25 USD per tick	addSpray() every 500ms with 0.25 USD equivalent
\$1.00 / second	0.50 USD per tick	addSpray() every 500ms with 0.50 USD equivalent
Custom	User-defined	UI converts any positive value to wei before call

SECTION 06

Notification Layer

On SprayClosed event emission, an off-chain indexer (The Graph subgraph or a lightweight FastAPI webhook listener) catches the event and:

- Resolves the honoree address to any registered display name (optional ENS / Celo Name Service).
- Pushes a notification payload via WebSocket to any open browser session watching that address.
- Logs the spray receipt in a Supabase table for history queries.

Notification format

```
"■ {sprayer_tag} sprayed ${total} on {honoree_name}!"
```

SECTION 07

Fee Model

SprayChain v1 charges zero protocol fees. 100% of sprayed value reaches the honoree. This is intentional — the product goal is adoption and cultural resonance, not extraction. Future versions may introduce an optional 0.5% tip to a community treasury, governed by SPRAY token holders.

SECTION 08

Privacy

- All spray transactions are on-chain and publicly visible.
- The honoree tag (name typed by sprayer) is stored as calldata — visible on-chain.
- No off-chain KYC or identity verification is required.
- SprayChain is non-custodial; FaucetDrops never holds user funds.

SECTION 09

Roadmap

Phase	Milestone
v1.0	Native CELO spray on Celo Mainnet + Botchain; HTML frontend; SprayMachine.sol
v1.1	cUSD / USDC ERC-20 spray; mobile-optimised PWA
v1.2	Party rooms — shared URL where multiple guests spray simultaneously; live leaderboard
v2.0	SPRAY governance token; community treasury; cross-chain bridges (Base, Arbitrum)
v2.1	NFT spray receipts — each spray session mints a commemorative NFT sent to honoree

SECTION 10

Conclusion

SprayChain is the first on-chain implementation of money spraying — a tradition that has connected communities, honored individuals, and expressed joy across Africa and the diaspora for generations. By encoding that ritual into a smart contract and giving it a cinematic frontend, SprayChain makes Web3 feel human, festive, and culturally rooted.

Build on SprayChain. Spray with pride.

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